

Integrating VirusTotal with Wazuh SIEM

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1 Introduction

Threat intelligence plays an essential role in modern Security Operations Centers (SOC) by providing context to security alerts and helping analysts distinguish between legitimate activity and potential threats. While traditional security monitoring identifies suspicious events, threat intelligence enriches these events with external reputation data, enabling faster and more accurate investigations.

VirusTotal is one of the most widely used threat intelligence platforms. It aggregates results from multiple antivirus engines and security vendors to determine whether a file, URL, domain, or IP address is malicious. By integrating VirusTotal with Wazuh, detected files can automatically be checked against VirusTotal's database, allowing Wazuh to enrich alerts with reputation information.

This project demonstrates the integration of VirusTotal with Wazuh in a controlled SOC laboratory. The integration was implemented using the official Wazuh documentation together with the VirusTotal API. After configuration, Wazuh successfully queried VirusTotal whenever monitored files were detected, providing additional threat intelligence directly within the Wazuh Dashboard.

2 About VirusTotal

VirusTotal is a cloud-based threat intelligence platform owned by Google that analyzes suspicious files, URLs, IP addresses, and domains using dozens of antivirus engines and security vendors.

VirusTotal provides:

- * File reputation analysis
- * URL reputation analysis
- * Domain reputation
- * IP reputation
- * File hashes (MD5, SHA1, SHA256)
- * Malware family identification
- * Community threat intelligence

Rather than acting as an antivirus solution, VirusTotal provides security analysts with additional information to support incident investigations.

3 Lab Environment

The VirusTotal integration was implemented using the following laboratory environment.

Host Machine

- * macOS (MacBook Air M2)

Virtualization Platform

- * UTM

Virtual Machines

- * Ubuntu Server

Purpose:

- * Wazuh Manager
- * Wazuh Dashboard
- * Wazuh Indexer

- * Windows 11

Purpose:

- * Wazuh Agent
- * File Integrity Monitoring
- * VirusTotal Integration

- * Kali Linux

Purpose:

- * Security Testing
- * Attack Simulation

4 Prerequisites

Before configuring the integration, the following requirements were completed:

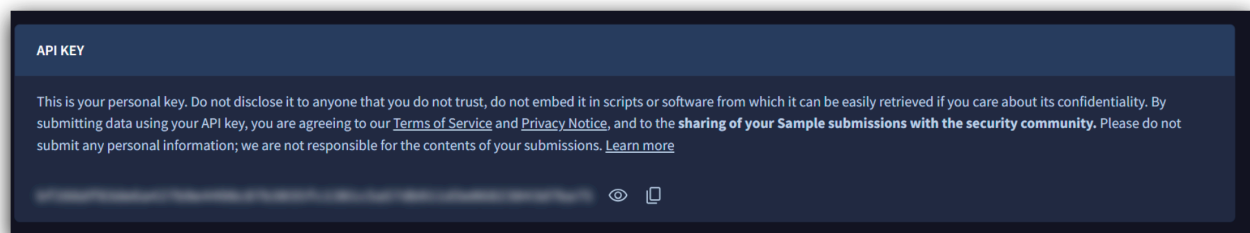
- * Wazuh Manager installed and operational
- * Windows Wazuh Agent successfully connected
- * File Integrity Monitoring configured
- * Active Internet connection
- * VirusTotal account
- * VirusTotal API Key

5 Obtaining the VirusTotal API Key

A VirusTotal account was created to obtain an API key for integration.

Steps performed:

1. Create a VirusTotal account.
2. Navigate to the user profile.
3. Open the API Key section.
4. Copy the personal API key.



The API key is required for Wazuh to communicate with the VirusTotal service.

6 Configuring the Wazuh Manager

The Wazuh Manager configuration was updated to enable VirusTotal integration.

The main configuration file was modified by adding the VirusTotal integration block.

Open C : \ Program Files (x86) \ ossec - agent \ ossec . conf
with a text editor (e . g . , Notepad) and add the c o n f i g u r a t i o n
snippet .

Insert the following block (replace YOUR API KEY with your actual key):

```
<!-- Virus Total Integration -->
<integration>
  <name>virustotal</name>
  <api_key>YOUR_API_KEY</api_key> <!-- Your API Key -->
  <group>syscheck</group>
  <alert_format>json</alert_format>
</integration>
```

```
<integration>
  <name> virustotal </name>
  <api_key>YOUR_API_KEY</api_key>
  <alert_format > json </alert_format>
</integration>
```

Save the file, then restart the Wazuh Manager to apply the changes.:

```
net stop wazuhsvc
net start wazuhsvc
```

7 Configuring the Wazuh Agent

The Windows Wazuh Agent was configured to monitor a directory using File Integrity Monitoring (FIM).

Open C : \ Program Files (x86) \ ossec - agent \ ossec . conf
with a text editor (e . g . , Notepad) and add the configuration
snippet.

Add the following Snippet:

```
<directories check_all="yes" realtime="yes">/home/virustotaltest</directories>
```

```
< syscheck >  
  < directories realtime = "yes" > / home / virustotaltest  
  </ directories >  
</ syscheck >
```

This ensures that any file added to /home/virustotaltest will trigger a VirusTotal lookup.

Restart the agent to apply changes:

```
net stop wazuhsvc  
net start wazuhsvc
```

8 Integration Workflow

The integration follows the sequence below:

1. A file is added to the monitored directory.
2. The Wazuh Agent detects the file through File Integrity Monitoring.
3. The file hash is generated automatically.
4. Wazuh sends the hash to VirusTotal using the configured API.
5. VirusTotal searches its malware database.
6. The reputation result is returned to Wazuh.
7. The enriched alert is displayed in the Wazuh Dashboard.

This process occurs automatically without requiring manual analysis.

9 Conclusion

This project successfully integrated VirusTotal with the Wazuh SIEM platform, enabling automatic threat intelligence enrichment for files detected through File Integrity Monitoring. By configuring the VirusTotal API within the Wazuh Manager and enabling real-time monitoring on the Windows endpoint, Wazuh was able to automatically submit file hashes for reputation analysis and display the results within the Wazuh Dashboard.

The integration enhanced the SOC laboratory by providing additional context for security alerts, allowing faster identification of malicious files and improving the overall incident investigation process. Combining Wazuh's monitoring capabilities with VirusTotal's extensive threat intelligence database creates a more effective and informative security monitoring solution for both learning and real-world SOC environments.